CONSISTENT QUANTIZATION OF A TWO-DIMENSIONAL NON-ABELIAN CHIRAL THEORY

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Received 30 March 1987

A two-dimensional SU(N) gauge model coupled to Weyl fermions is studied following recent suggestions for the quantization of potentially anomalous chiral theories. The Weyl fermion determinant is evaluated and the fermionic current is shown to be conserved due to the gauge invariance of the resulting quantum theory As in the abelian case, the vector meson acquires a mass and the model is consistent provided a regularization parameter is conveniently chosen.

The possibility of consistently quantizing theories in which gauge and gravitational symmetries of Weyl fermions are affected by anomalies [#1] has been independently proposed by a number of authors [2–5] after Faddeev and Shatashvili's [6] suggestion concerning the modification of the original classical lagrangian by the introduction of a new *physical* (chiral) field with a Wess–Zumino action, this rendering the quantum theory gauge-invariant.

Indeed, it has been shown in refs. [2–4] that a proper quantum treatment of the gauge degrees of freedom in chiral gauge theories (as well as in gravitational ones [5]) uncovers the presence of the Wess–Zumino term exactly as it happens with the Liouville action in the quantization of the string, as first shown by Polyakov [7]. In both cases, some degrees of freedom, classically decoupled, reappear as physical after quantization, this automatically ensuring gauge invariance.

It has to be stressed that the above-mentioned results were derived within the lagrangian path-integral framework, assuming as usually [8] that integration "over all fields" leads to a correct quantum theory. An analysis within the canonical hamiltonian formalism is still missing (see, however, refs. [9,10]). There are, however, many indications, particularly in detailed analysis of two-dimensional abelian models [11–14] and gravitational theories [5,15] suggesting the consistency of the new proposal. Of course, unitarity and renormalizability of the resulting theory have to be re-examined but, since it is the loss of gauge invariance which is usually thought at the root of unrenormalizability and non-unitarity, one can hope (and in fact there are some indications, see ref. [16]) that the absence of anomalies should help in making the theory consistent.

Although these problems have to be analysed in four-dimensional theories, the study of simpler two-dimensional models sheds some light on possible issues (at least on unitarity and positivity questions, as first shown by Jackiw and Rajaraman for the chiral Schwinger model (CSM) [11]). It is the purpose of this note to extend the analysis of the CSM [9–14,17] to the non-abelian case [15] using the proposal of refs. [2–4].

We shall see that as in the Dirac fermion case [18] the Weyl fermion determinant can be exactly evaluated leading to a non-linear sigma model with Wess–Zumino term [19,20]. The resulting effective lagrangian has a regularization dependence (on the a-parameter introduced by Jackiw and Rajaraman [11]) which plays a central role in the consistency of the model. After computing the fermion determinant we analyse the complete effective model showing that for certain values of a ($a>1$) the theory can be defined in a unitary positive definite way, gluons become massive as in the Dirac fermion case [21] and the fermion current is conserved, this lead-

[#1] See ref. [1] for an updated review.

ing to a gauge invariant theory with non-anomalous commutators. Although these results strongly rely on the fact the model is two-dimensional, we think the success in quantizing it consistently makes reasonable to hope that four-dimensional non-abelian theories can be handled in a similar way.

We start from the two-dimensional (euclidean) lagrangian:

$$\mathscr{L}=-\tfrac{1}{2}\operatorname{tr}F_{\mu\nu}^2+\bar{\psi}\mathrm{D}(A)\psi, \tag{1}$$

$$\mathrm{D}(A)=(\mathrm{i}\partial\!\!\!/+eA\!\!\!/)\tfrac{1}{2}(1-\gamma_5), \tag{2}$$

where the gauge field A_μ takes values in the Lie algebra of SU(N) and the Weyl fermions are taken in the fundamental representation of SU(N).

In the lagrangian formulation of the path-integral quantization one should consider the generating functional Z as an integral over the whole space of connections rather than as an integral over the orbit space [22]. Although there is no harm in looking at Z as an integral over orbits when Dirac fermions are present, the situation radically changes for Weyl ones: their ability to create gauge non-invariant interactions prevents the usual factorization of the gauge-group volume [2]. We define Z as

$$Z=\int \mathscr{D}\bar{\psi}\,\mathscr{D}\psi\,\mathscr{D}A_\mu \exp\left(-\int \mathscr{L}\mathrm{d}^2x\right), \tag{3}$$

and use the Faddeev–Popov resolution of the identity as a convenient choice of the integration variables. The resulting generating functional, as first derived in refs. [2–4] reads

$$Z=\int \mathscr{D}\bar{\psi}\,\mathscr{D}\psi\,\mathscr{D}A_\mu\delta(F[A])\Delta_{\mathrm{FP}}(A)$$

$$\times\,\mathscr{D}gJ(g^{-1},A^g)\exp\left(-\int \mathscr{L}\mathrm{d}^2x\right), \tag{4}$$

with

$$A_\mu^g=g^{-1}A_\mu g+(\mathrm{i}/e)g^{-1}\partial_\mu g, \tag{5}$$

and $F[A]$ some gauge condition leading to a Faddeev–Popov determinant $\Delta_{\mathrm{FP}}(A)$. The presence of a non-trivial jacobian J prevents the factorization of the gauge-group volume,

$$J(g,A)\equiv\det\mathrm{D}(A)/\det\mathrm{D}(A^g), \tag{6}$$

since $J\neq 1$ for Weyl fermions [2–4]. We then define an effective action by integration over fermions and *over the g-field*:

$$\exp(-S_{\mathrm{eff}}[A])=\int \mathscr{D}\bar{\psi}\,\mathscr{D}\psi\,\mathscr{D}g$$

$$\times\exp\left(-\int\bar{\psi}\mathrm{D}(A)\psi\mathrm{d}^2x\right)J(g^{-1},A). \tag{7}$$

This effective action satisfies the important relation

$$S_{\mathrm{eff}}[A^h]=S_{\mathrm{eff}}[A], \tag{8}$$

i.e. it is a gauge-invariant effective action in the background of A_μ (it plays the role of the fermion determinant in ordinary gauge invariant theories with Dirac fermions). Relation (8) translates into BRS invariance of the full theory [2]. From (7), VEV's of currents and current algebra can be evaluated, following the steps described in ref. [17] for the CSM.

Our first task is the evaluation of the fermion determinant. The first problem to handle is related to the fact that the Dirac operator (2) does not have a well-defined eigenvalue problem. As usually, we add right-handed free fermions defining a new operator with a well-posed eigenvalue problem:

$$\hat{\mathrm{D}}=\mathrm{D}(A)+\mathrm{i}\partial\!\!\!/\tfrac{1}{2}(1+\gamma_5). \tag{9}$$

Then, up to a normalization constant we have [23]

$$\det\mathrm{D}(A)\equiv\det\hat{\mathrm{D}}|_{\mathrm{Reg}}, \tag{10}$$

where we indicate with "Reg" the necessity of a regularization of the $\hat{\mathrm{D}}$'s eigenvalues product (which otherwise grows without bound). Now, the point is that $\hat{\mathrm{D}}$ is not covariant and hence it does not impose itself as the natural regulator in a heat-kernel regularization scheme. We can then use a more general regulator $\mathrm{D}_{\mathrm{Reg}}$ introduced in the heat-kernel approach through the identity

$$1=\lim_{M^2\to\infty}\exp(-\mathrm{D}_{\mathrm{Reg}}^2/M^2),$$

$$\mathrm{D}_{\mathrm{Reg}}=\hat{\mathrm{D}}(A)+\tfrac{1}{2}aeA\!\!\!/(1+\gamma_5), \tag{11}$$

with a the Jackiw–Rajaraman [11] parameter. We are now in conditions of evaluating the Weyl determinant using eqs. (10), (11). Following ref. [24] one can easily show that

$$\log\det \mathrm{D}(A) = -(1+a)\frac{e^2}{16\pi}\int \mathrm{d}^2x\,\mathrm{tr}\,A^2$$
$$-\frac{e^2}{\pi}\int_0^1 \mathrm{d}t \int \mathrm{d}^2x\,\mathrm{tr}[\gamma_5\phi A_t A_t + \mathrm{i}\gamma_5\eta A_t A_t$$
$$-(1/e)\gamma_5 \not\partial\eta A_t - (1/e))\not\partial\eta A_t], \tag{12}$$

where $A_{t=1}$ corresponds to the gauge field A expressed in terms of fields ϕ, η taking values in the Lie algebra of SU(N). As it is to be expected in a two-dimensional model, the regularization ambiguity can only affect quadratic terms in A_μ [25]. This is the reason why the Wess–Zumino term (which can be identified as the first term in the parentheses in (12) for the decoupling $\eta=0$ gauge [21]) has an a-independent coefficient. Indeed, in the decoupling gauge (12) reads

$$\log\det \mathrm{D} = (1+a)\frac{e^2}{8\pi}\int \mathrm{d}^2x\,\mathrm{tr}\,\partial_\mu U^{-1}\partial_\mu U$$
$$+\frac{\mathrm{i}}{4\pi}\int_0^1 \mathrm{d}t\int \mathrm{d}^2x\,\epsilon_{\mu\nu}\,\mathrm{tr}(U_t^{-1}\partial_\mu U_t U_t^{-1}$$
$$\times\partial_\nu U_t U_t^{-1}\partial_t U), \tag{13}$$

with $U_t = \exp(2t\phi)$, $U = U_1$.

One can easily write the effective action in the decoupling gauge by identifying $\exp(\mathrm{i}\eta) = g$:

$$S_{\mathrm{eff}}[A] = \int \mathscr{D}g \det \mathrm{D}(A^{g=\mathrm{e}^{\mathrm{i}\eta}}). \tag{14}$$

Although in the present non-abelian case the analysis of the effective lagrangian (which results of adding $\mathscr{L}_{\mathrm{eff}}$ and the $F^2_{\mu\nu}$ term) is more involved than in the U(1) case, one can just consider the quadratic terms in ϕ and η to infer if gluons get a mass through the Schwinger mechanism. The resulting gaussian integral over η can then be performed and the result is that the ϕ-field gets a mass

$$m^2 = (e^2/8\pi)a^2(a-1), \tag{15}$$

exactly as in the Dirac fermion case [21]. We then see that in the non-abelian chiral model we get the same result as in the (chiral) Schwinger model case [11,17]: for $a>1$ the model can be defined in a unitary, positive definite way (it can be shown that this is also the case for the limiting $a=1$ case [26,1]). There are N^2-1 massive particles (the ϕ-fields) with a mass lying everywhere between $e^2/2\pi$ and infinity, in contrast with the conventional (Dirac fermionic) model where gauge invariance fixes the boson mass (precisely to $e^2/2\pi$, see ref. [21]). It is only in the $a<1$ region where unitarity is lost. There is no satisfactory explanation of why only a range of the undetermined parameter ($a\geqslant 1$) gives a unitary positive definite model and if a similar behavior is to be attended in four-dimensional models where also renormalizability has to be analysed.

Concerning the fermionic current, following ref. [17] we define

$$J_\mu \equiv -\delta S_{\mathrm{eff}}/\delta A_\mu, \tag{16}$$

which, due to eq. (8) is automatically conserved:

$$\mathrm{D}_\mu J^\mu = 0. \tag{17}$$

After some calculations, one gets for J_μ (written in terms of lightcone components $J_\pm = J_0 \pm \mathrm{i}J_1$)

$$J_\pm = (-\mathrm{i}/4\pi)U^{\pm 1}\mathrm{D}_\pm U^{\mp 1}$$
$$-\frac{ea}{4\pi}\frac{1}{Z}\int \mathscr{D}g \det \mathrm{D}(A^g)\, gA^g g^{-1}. \tag{18}$$

It is important to stress that if the jacobian J was not included in eq. (4), the effective action just coincides with the logarithm of the fermion determinant (as can be trivially seen from eq. (7) making $J=1$). Since there is no possible definition of this determinant in a gauge-invariant way [27], the resulting current would be anomalous. It is the integration over g in eq. (7) which ensures gauge invariance of the effective action and hence non-anomalous currents.

It is also interesting to note that the first term in (18) coincides with the currents obtained in the conventional (Dirac fermionic) non-abelian model while the second term represents another manifestation of the regularization ambiguities in chiral models. Its role can be clarified studying the current algebra which can be derived from eq. (14) as explained in ref. [17] using the Bjorken–Johnson–Low procedure. In the present non-abelian case, the calculation is more complicated but due to eqs. (8) or (17), (18) one necessarily has to arrive to non-anomalous (but a-dependent) commutators. We hope to report on this subject in a forthcoming publication.

Note added. After completing this investigation, we became aware of ref. [28] where the chiral determinant (eq. (12)) is also evaluated.

We would like to thank Carlos Naón for numerous helpful discussions. M.V.M. and M.T. are partially supported by CONICET. F.A.S. is partially supported by CIC, Buenos Aires, Argentina.

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